

Subjectivity-Intersection Mathematics

A Mathematical Research Program for Viewpoints, Relational Generation, and Simultaneous Closure

Research White Paper - English Edition

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Abstract

Subjectivity-Intersection Mathematics is a mathematical research program for describing what occurs when multiple subjects approach one another's viewpoints while preserving their differences: transformations between viewpoints, information lost in those transformations, a third state generated by the relation, and the action of that state back upon both sides. Its canonical structure is written as " $AB \rightarrow C \rightarrow AB$." A and B are two viewpoint states representing distinct subjects; C is a state generated by their relation; and the second arrow denotes the action of C back upon the relation between A and B.

Celestial motion provides a precise and reproducible test model. The program has examined observer-dependent phenomena, information loss under viewpoint transformation, joint constraints among three viewpoints, noncommutative operators, history-dependent states, three-state closure, and predictive coupling between viewpoint series. These studies show that a mathematics of viewpoints must distinguish at least three realizations of C: reconstruction of an existing object, a common coordinate transformation, and a state generated by the relation between viewpoints.

On the theoretical side, a restricted no-go result shows that a memoryless, continuous, exchange-symmetric, orthogonally equivariant rule mapping two unoriented bearings to a third bearing can produce only an angle bisector or its perpendicular. An irreducible third state therefore requires history, orientation, an independent relational state, or a genuine third viewpoint. A three-state dynamics produced a stable non-consensus closure with the three states separated by 60 degrees. A null audit then showed that 20,000 random initial triads converged to the same closure with a 100% rate, identifying it as an attractor generated by the chosen law rather than a celestial-specific structure.

The clearest empirical discovery is that a time-dependent coordinate gauge can generate cross-predictive gain that is indistinguishable from directed physical coupling at the level of the tested statistic. The held-out R-squared gain first found for Venus replicated over 1965-1999, 2000-2035, and 2035-2070 under two computational methods, but vanished in J2000 fixed axes and when nutation was removed. Reinjection of the extracted common nutation angle restored the gain. Across six non-overlapping windows spanning 1895-2105, the result was 6/6 with nutation, 0/6 without nutation, and 6/6 after reinjection.

Subjectivity-Intersection Mathematics must therefore do more than place viewpoints inside mathematics. It must distinguish relational generation from reconstruction of existing objects, common coordinate motion, law-induced attractors, and analysis-induced spurious coupling. This white paper integrates the conceptual classification, mathematical models, counterexamples, and validation procedure required for that task.

Principal Results

- 1 Planetary retrograde motion was formulated as an orientation reversal jointly generated by Earth, a planet, and an observation operator.
- 2 A round trip between viewpoints closes when the complete relational state is retained; projection to angle alone removes a distance fiber and leaves information holonomy.
- 3 A distance-ratio carrier, unique up to common scale, was recovered from the joint constraint among the three Sun-Earth-planet directions.
- 4 A decision framework was constructed that distinguishes C as objective reconstruction, common gauge, or a candidate for relational generation.
- 5 A restricted no-go result was obtained for constructing an irreducible third state from the instantaneous values of two viewpoints alone.
- 6 A history lift produced Z2 branch holonomy that cannot be reduced to the instantaneous state.
- 7 A stable non-consensus three-state closure was constructed, while a null audit attributed that closure to the update law rather than to celestial initial conditions.
- 8 A common nutation gauge was shown to generate reproducible spurious cross-predictive gain through fixed-frame controls, component removal, reinjection, a full planetary panel, and six-window replication.

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1. Why Subjectivity-Intersection Mathematics Is Needed

Modern science developed by extracting quantities that remain invariant across observers. This objectifying operation is indispensable. Yet when observational position, body, instrument, coordinate system, or readout rule participates in the constitution of a phenomenon, treating all of them as error can remove the relational structure that makes the phenomenon possible.

Planetary retrograde motion is a simple example. An outer planet may remain prograde in heliocentric coordinates while its geocentric longitude temporarily reverses direction. Retrograde motion is not a property of the planet alone; it arises from the relation among Earth, the planet, and the observation operator. It is observer-dependent while remaining calculable, predictable, and shareable among observers.

This example requires a distinction between observer dependence and arbitrariness. Subjectivity-Intersection Mathematics describes, as structure, what is preserved, what is lost, and what is generated between viewpoints under specified projections.

Its central question is:

When two distinct viewpoints A and B intersect while preserving their differences, can an irreducible relational state C arise? If so, how does C act back upon A and B and close the whole of A, B, and C?

2. Basic Concepts

2.1 Subject and Viewpoint

A subject is the side on which experience, observation, and meaning take place. Mathematics does not identify the subject itself with a coordinate or number. It works with the viewpoints, expressions, actions, and relational changes through which a subject reads the world.

A viewpoint is the mathematical representation of the position, projection, or operation through which a subject reads an object. Different viewpoints of the same object can produce different coordinates, directions, visibility, and information. Subject and viewpoint are therefore distinct: the subject is the side on which observation occurs; the viewpoint is the representation connecting that activity to mathematics.

2.2 Three Subjective Poles

Concept	Meaning	Connection to mathematics
Absolute subjectivity	The originating subjective pole by virtue of which individual viewpoints can arise	An ontological hypothesis that multiple projections can arise
Relative subjectivity	An individual experiencing subject with a body, place, time, and history	Modeled through viewpoint states, projections, expressions, and actions
Intersection subjectivity	An irreducible relational pole arising with the intersection of multiple relative subjects	Investigated through observable consequences of a third state and closure

Intersection subjectivity is not an average or fusion of two subjects. It is a third relational pole that arises when both sides preserve their differences and retain the other's viewpoint within the relation, rather than assimilating one side to the other.

2.3 Intersection

Intersection does not mean that two states become equal. It begins when a one-way reading in which A objectifies B only from A's viewpoint is exceeded, and reciprocal viewpoint transformation generates a relational state while preserving the difference carried by B's viewpoint.

Intersection has four elements:

- 1 A and B preserve distinct viewpoints.

- 2 Relational information is generated that cannot be obtained from either side alone.
- 3 The generated relational state acts back upon both A and B.
- 4 A, B, and the relational state close as mutual conditions.

2.4 Theoretical Position

Subjectivity-Intersection Mathematics builds on second-order cybernetics, which places the observer inside the system, and on the formalization of distinction, re-entry, and self-reference in Laws of Form (Spencer-Brown 1969; Kauffman 1995). It extends the problem from the self-inclusion of one observer to reciprocal transformations among multiple viewpoints and the generation of a difference-preserving third state.

Leighton and Hankey (2026) provide a closely related construction in which reciprocal dual structure survives iterative agreement closure and stability requires a role-separated witness rather than a bare two-agent exchange. This gives a direct formal precedent for preserving difference within closure while leaving the present criteria for a relation-generated third state to be tested.

The relevant structures are fibers lost under projection, holonomy remaining after a closed path, noncommutativity recording order of action, feedback regenerating a relation, and simultaneous closure establishing several states as mutual conditions. The program connects them as one problem of relational generation.

3. Canonical Mathematical Structure

3.1 $AB \rightarrow C \rightarrow AB$

For each model, let the state spaces of A and B be X_A and X_B , the relational state space be X_C , and let H represent history or context. Define relational generation by $G: X_A \times X_B \times H \rightarrow X_C$ and feedback by $F_A: X_A \times X_C \rightarrow X_A$ and $F_B: X_B \times X_C \rightarrow X_B$.

" $AB \rightarrow C \rightarrow AB$ " means that $C = G(A, B, H)$ is generated, $A' = F_A(A, C)$ and $B' = F_B(B, C)$ are formed, and C is again generated consistently from the updated relation. The required properties are joint generation of C, preservation of the difference between A and B, reciprocal action through C, and recursive consistency.

3.2 What C Represents

C need not be the same type of object in every model. It may be a distance ratio or joint kernel in a geometric model, a commutator or feedback generator in an operator model, a sheet on a covering space in a history model, or a statistic representing a relation in a time-series model. What is common is its functional role: it is generated by the relation between A and B and constrains that relation.

Class	Functional meaning	Criterion
C_objective	Multiple viewpoints jointly reconstruct an existing object	C coincides with an object existing before the intersection
C_gauge	A common representational transformation is applied to A and B	Relational invariants are preserved although A and B change
C_intersection candidate	A state generated by the relation changes that relation and closes recursively	Difference is preserved, gauge-invariant relational quantities change, and closure occurs

3.3 Criteria for a C_intersection Candidate

- 1 Generation of C requires both A and B.
- 2 C retains information about the difference between A and B.

- 3 A and B remain distinct after the action of C.
- 4 The action of C changes a gauge-invariant relational quantity.
- 5 A, B, and C close recursively as mutual conditions.
- 6 The construction gives consequences distinct from reconstruction, averaging, consensus, or common intersection.

Mathematical C is a model of the structure under which intersection subjectivity may arise, rather than a direct representation of intersection subjectivity itself. Extracting conditions shared across distinct models and connecting them to observable consequences is the central task of the program.

4. Why Celestial Motion Is Used as a Test Model

Celestial motion has three advantages as a test bed. Heliocentric and geocentric viewpoints are explicit; the same physical system generates distinct observational series with high precision; and coordinate frames, dynamical approximations, full ephemerides, and independent periods can be varied to falsify hypotheses.

The role of the celestial experiments is to test the operator structures required by Subjectivity-Intersection Mathematics in a system whose coordinates and motion are well characterized.

Experiment	A and B	Realization of C	Condition tested
Planetary retrograde	Heliocentric description and geocentric observation	Orientation reversal generated by observation	Reality and reproducibility of viewpoint-dependent quantities
Viewpoint round trip	Heliocentric and geocentric states	Complete relational state and lost fiber	Relational information required for viewpoint transformation
Three-viewpoint closure	Three directed edges	Positive joint kernel and distance ratios	Joint constraint preserving partial viewpoints
Noncommutative action	Two projection operators	Fixed point or commutator	Distinguishing C_objective from C_gauge
History lift	Two unoriented bearings	History sheet on a covering space	Relational state exceeding instantaneous values
Three-state dynamics	Three rank-one projectors	60-degree tight frame	Non-consensus closure and law-induced attractor
Predictive test	Heliocentric and geocentric series	Held-out predictive gain	Distinguishing physical relation from common coordinate gauge

5. Observer-Dependent Phenomena and Relational Carriers

5.1 Planetary Retrograde

Let z_E and z_P be the heliocentric positions of Earth and a planet, and define the geocentric direction by $\text{Pi}_E(z_E, z_P) = \arg(z_P - z_E)$. Even when the heliocentric longitude velocity of an outer planet is positive, the derivative of this observable can become negative. Neighborhoods of stationary points fit a local quadratic fold with high accuracy.

Retrograde motion is therefore an orientation reversal generated by Earth, the planet, and the observation operator. Explaining its cause in heliocentric dynamics does not remove the phenomenon constituted in the geocentric viewpoint.

5.2 Viewpoint Round Trips and Fibers

Let E be Earth's heliocentric position, P the planet's heliocentric position, and $Q = P - E$ the geocentric relation. If the complete relational state (E, P, Q) is retained, a round trip between heliocentric and geocentric viewpoints closes.

Projection to longitude alone removes distance information. Multiple distance states correspond to the same longitude, so the inverse transformation becomes a multivalued correspondence. If the missing distance is supplied by an arbitrary convention, the round trip does not return to the initial state. The residual is information holonomy.

5.3 Joint Constraint Among Three Viewpoints

Let u_1 , u_2 , and u_3 denote the three directions joining Sun, Earth, and planet. When positive coefficients ρ_1 , ρ_2 , and ρ_3 satisfy $\rho_1 u_1 + \rho_2 u_2 + \rho_3 u_3 = 0$, a nondegenerate planar configuration determines the three distance ratios uniquely up to common scale.

Every observed time-aligned configuration had a positive joint kernel. When time correspondence was destroyed, the rate fell to approximately 25-44%. The distance-ratio carrier was recovered from the joint constraint without averaging or changing the directions. This provides a minimal geometric model of a relational carrier.

6. Noncommutativity, History, and the Third State

6.1 Noncommutativity Is Not Sufficient

Sequential composition of projections associated with different viewpoints generally depends on order. If their common fixed point is an existing target, however, it is $C_{\text{objective}}$. Noncommutativity records viewpoint difference but does not guarantee an irreducible third state.

For projection operators A and B , the commutator $C = [A, B] = AB - BA$ cannot be generated from A or B alone. Yet conjugating both by the same $R = \exp(\epsilon C)$ changes them while preserving their relative angle and distance. This realization is C_{gauge} .

6.2 Opposed Feedback

Returning C to A and B in opposite directions changes gauge-invariant relational quantities while preserving their difference. Under iteration, however, the dynamics converged to orthogonality, the state of maximal difference, rather than to a stable internal third state. The result is an eigenform of differentiation.

6.3 Two-Viewpoint No-Go Result

Consider a rule that maps two unoriented bearings to a third bearing and is memoryless, continuous, exchange-symmetric, and $O(2)$ -equivariant. Normalize the inputs to $\{-\delta/2, +\delta/2\}$. Reflection exchanges the inputs. Exchange symmetry and reflection equivariance then restrict the output locally to the angle bisector or its perpendicular.

A numerical symmetry audit over 100,000 random pairs with seed 20260719 supported this classification. Under these assumptions, an instantaneous two-viewpoint rule cannot escape the consensus center or its dual.

6.4 History Lift

Continuously lifting unoriented angle through time and tracking a branch of the bisector makes C a path-dependent state rather than a function of the instantaneous pair. Around a loop enclosing a degeneracy, the input line pair returns to the same state while C returns on the perpendicular sheet, producing Z_2 branch holonomy.

History exceeds the memoryless no-go assumptions and extends the third state from an instantaneous function to a path-dependent state.

7. Three-State Closure and Null Audit

The history state C was promoted to the same status as A and B, and all three were updated under a symmetric frame-potential gradient. The closure condition is $P_A + P_B + P_C = (3/2)I$, a tight frame of three unoriented lines separated by 60 degrees.

Celestial initial states converged to this non-consensus closure. The same occurred for 20,000 random initial triads and for history-destroyed C, with a 100% closure rate at learning rates 0.05, 0.10, and 0.20. Zero feedback did not close, and reversing the sign produced a different attractive state.

The experiment establishes a stable non-consensus closure for this three-state dynamics. The null audit attributes the 60-degree geometry to the selected update law rather than to celestial configuration or history. Data entering an attractor of a law must be distinguished from data selecting that structure.

8. Discovery of the Time-Dependent Coordinate Gauge

8.1 Initial Discovery

The history of heliocentric bearing was added to a baseline using only the target's geocentric bearing history, and held-out R-squared gain was measured. Sampling was every five days, history was 30 days with lags 1-6, the train-test split was 70/30, and the null distribution used period-preserving circular shifts.

For Venus under Swiss Ephemeris, gain was 6.818×10^{-7} in 1965-1999, 6.719×10^{-7} in 2000-2035, and 8.272×10^{-7} in 2035-2070. Under Moshier the corresponding values were 6.792×10^{-7} , 6.728×10^{-7} , and 8.306×10^{-7} . Exact p-values were approximately 0.000400 in all conditions.

8.2 Dynamical Model Decomposition

The effect did not replicate in fixed circular, fixed elliptical, or piecewise Kepler models with windows of 90 days or longer. A DOP853 point-mass Newtonian system from Mercury through Neptune converged numerically, with bearing differences of $3.35\text{--}5.06 \times 10^{-12}$ rad, maximum energy drift of 2.70×10^{-14} , and maximum angular-momentum drift of 1.13×10^{-14} , but gain was negative in all three periods.

The contrast between full ephemerides and the Newtonian model initially suggested a perturbative physical carrier. Fixing the coordinate frame changed that interpretation.

8.3 Frame Removal

Of-date coordinates were positive in all three periods, while the result vanished in fixed J2000 axes. Removing nutation with NONUT while retaining of-date coordinates reduced gain to a Swiss median of approximately -2.1×10^{-10} and a Moshier median of approximately -9.0×10^{-10} . TRUEPOS had little effect.

8.4 Reinjection of the Common Nutation Angle

Nutation angles extracted independently from Earth and Venus differed by at most 5.68×10^{-14} rad and were identical to numerical precision. Their peak-to-peak amplitude was approximately 1.83×10^{-4} rad.

Reinjecting only 25% of the angle into the NONUT series was enough to pass the three-period criterion; reinjecting 100% restored the original of-date gain exactly. Reversing the sign also produced gain. The sufficient condition was therefore a common nonlinear time-dependent rotation, not a directional physical action.

8.5 Planetary Panel and Long-Window Replication

In of-date coordinates, all seven planets except Mercury were consistently positive across the three periods. No planet was consistently positive across all three periods under NONUT or J2000. This rejected the initial Venus-specific interpretation.

The years 1895-2105 were then divided into six non-overlapping 35-year windows. Under both Swiss and Moshier, of-date was positive in 6/6 windows, NONUT in 0/6, and nutation reinjection in 6/6.

Swiss and Moshier use different computational methods but are not independent physical theories at the level of coordinate conventions. Their agreement establishes replication across numerical backends, while independence from the common coordinate gauge is supplied by the J2000, NONUT, and reinjection controls.

9. Mathematical Meaning of the Discovery

Let the underlying angle series be a_t and b_t , and let γ_t be a common coordinate rotation. The observed series are $a'_t = a_t + \gamma_t$ and $b'_t = b_t + \gamma_t$. Their increments share $\Delta \gamma_t$.

A self-history baseline uses only past $\Delta b'_t$. A cross-history model also uses past $\Delta a'_t$ and can therefore estimate the common component from two different mixtures. Predictive gain can arise without a directed $a \rightarrow b$ action.

A null that circularly shifts only the predictor destroys the simultaneity of the common gauge. The test can therefore assign a small p-value to gauge simultaneity rather than to physical action. Reproducibility, significance, and agreement across backends do not guarantee physical causality when the coordinate transformation is shared.

This is a constructive celestial-ephemeris counterexample showing that cross-history predictive gain is not invariant under time-dependent coordinate transformations.

10. Validation Principles for Subjectivity Intersection

When common structure or predictive relation appears between viewpoints, investigate it in the following order:

- 1 Exclude numerical error, boundary handling, and implementation differences.
- 2 Determine whether an existing object has merely been reconstructed.
- 3 Fix or remove coordinate motion common to the viewpoints.
- 4 Use random initial conditions to test whether the update law generates the result automatically.
- 5 Use surrogates to test whether the analysis creates spurious coupling from common signals or temporal alignment.
- 6 After removal, reinject the candidate component and test whether the result is restored.
- 7 Apply the fixed procedure to unused periods, alternative backends, and alternative targets.
- 8 Test whether a gauge-invariant relational quantity changes.

Removal alone does not identify a cause. Reinjection must reconstruct the result. A structure that necessarily appears under random controls should be reported as a property of the law rather than as a discovery in the data.

11. Established Results and Open Problems

11.1 Established Results

- Observer-dependent phenomena can be formalized as relations including the observation operator.
- Complete-state viewpoint transformation can be distinguished from projected information loss through fibers and holonomy.
- Common fixed points of noncommutative operators can be classified as $C_{\text{objective}}$ and common conjugation as C_{gauge} .
- A restricted no-go result holds for the memoryless two-viewpoint model.
- A history lift generates a path-dependent third state.

- A stable non-consensus closure can be constructed in a three-state dynamics.
- Law-induced attractors can be identified through random controls.
- A common time-dependent coordinate gauge can produce reproducible predictive gain between viewpoint series.
- Frame removal and reinjection can identify the source of that spurious coupling.

11.2 Open Problems

- Axiomatize $C_{\text{intersection}}$ and establish necessary and sufficient conditions.
- Determine the minimal combination of history, orientation, independent relational state, and third viewpoint required for an irreducible C .
- Construct predictive statistics invariant under common time-dependent coordinate transformations.
- Define data-dependent quantities in three-state closure that are not guaranteed by the update law.
- Transfer the coordinate-audit method to non-celestial systems with reciprocal response and relational history.
- State the additional hypotheses connecting operator-observable correspondences to the ontology of subjectivity intersection.

12. Research Novelty

The program's novelty lies in integrating viewpoints, noncommutativity, history, closure, and gauge into a single falsification sequence for relational generation.

First, it distinguishes $C_{\text{objective}}$, C_{gauge} , and $C_{\text{intersection}}$ candidates rather than treating noncommutativity or a fixed point as sufficient evidence for a third state.

Second, it treats the two-viewpoint no-go result, history lift, three-state closure, and audit of law-induced attractors as consecutive parts of one constructive problem.

Third, it demonstrates that the standard astronomical coordinate transformation of nutation can produce spurious predictive gain replicating across independent periods and two backends, and traces the effect through fixed-frame disappearance, NONUT disappearance, extracted-angle reinjection, a planetary panel, and six-window confirmation.

Common signals and latent confounding in predictive-causality analysis are known problems. The specific new contribution is the reproducible constructive counterexample provided by a time-dependent astronomical coordinate gauge and the diagnostic sequence used to identify it.

13. Research Program

13.1 Formal Conditions for $C_{\text{intersection}}$

Axiomatize joint generation from A and B, irreducibility to either input, distinction from common gauge, change in relational invariants under feedback, preservation of difference, and recursive closure. Establish independence among necessary conditions and candidates for sufficiency.

13.2 Gauge-Invariant Predictive Relations

Redesign prediction tests using relative quantities, curvature, holonomy, and quotient-space quantities preserved under common rotation. Establish invariance under common time-dependent transformations.

13.3 History and Independent Relational State

Formalize the history sheet as an independent relational state rather than auxiliary memory, and determine the minimal state dimension required to exceed the two-viewpoint no-go result. Seek stability conditions preventing the third state from collapsing to consensus, its orthogonal dual, or maximal difference.

13.4 Non-Celestial Data

Celestial models are well suited to coordinate audits but do not directly measure reciprocal understanding or co-generation. The next stage concerns systems in which two agents respond to one another and relational history is recorded independently. The target observable should require the relational state for its explanation and remain unpredictable from the individual states alone.

14. Conclusion

Subjectivity-Intersection Mathematics describes what is preserved, lost, reconstructed as an existing object, expressed as common coordinate motion, and generated relationally among multiple viewpoints.

The celestial experiments instantiate observer-dependent phenomena, information fibers, joint-constraint carriers, noncommutative action, history holonomy, and non-consensus closure. They further classify apparent third states as existing objects, common conjugations, law-induced attractors, time-dependent coordinate gauges, or candidates for relational generation.

The program has established mathematical boundaries separating candidates for C from existing objects, gauges, update laws, and analysis artifacts. Its next stage is to formalize C_intersection and test non-celestial interactive systems with relational quantities invariant under common coordinate transformations.

Appendix A. Terminology

Term	Meaning
Subject	The side on which experience, observation, and meaning arise
Viewpoint	A position, projection, or operation through which an object is read
Intersection	Difference-preserving joint generation of a relational state that acts back upon both viewpoints
Fiber	Degrees of freedom lost under projection and corresponding to the same observed value
Holonomy	State change remaining after traversal of a closed path
Relational carrier	Relational information that jointly sustains multiple viewpoints or states
Simultaneous closure	A, B, and C arising as mutual conditions rather than in a one-way sequence
Gauge	Freedom of coordinates or representation that leaves object relations unchanged
C_objective	Reconstruction of an existing external object
C_gauge	A common transformation preserving relational invariants
C_intersection candidate	An irreducible third state generated by a relation and acting back upon it
Eigenform	A form stably regenerated through recursive action

Appendix B. Experimental Conditions

Predictive-Gain Test

Item	Condition
Sampling interval	5 days
Bearing	Unoriented angle
History	30 days, lags 1-6
Model	OLS
Split	70% training, 30% evaluation
Baseline	Target self-history
Full model	Self-history plus other-viewpoint history
Endpoint	Held-out R-squared gain
Null	All admissible circular shifts excluding shifts below 30 samples
Period criterion	Positive gain and $p < 0.0166667$

Three-State Closure Audit

Item	Condition
Random samples	20,000 triads
Random seed	20260719
Maximum iterations	1,000
Learning rates	0.05, 0.10, 0.20
Closure condition	A tight frame of three unoriented lines separated by 60 degrees

Numerical Audit of the Two-Viewpoint No-Go Result

Item	Condition
Random samples	100,000 pairs
Random seed	20260719
Audit targets	Exchange, rotation, reflection, and unoriented-representation symmetries

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